

UML Use Case Diagrams

Definition

A UML use case diagram is a **graphical representation** of a system's use cases, actors, and their relationships. It provides a **black-box view** of the system and is particularly useful during the early phases of software development.

Purpose and Benefits

- Illustrates the names of **use cases** and **actors**.
- Shows relationships such as *associations*, *includes*, and *extends*.
- Organizes use cases to improve **communication** and reduce **duplication** of text.
- Provides a **black-box perspective**—shows what the system does, not how it does it.
- Particularly useful during **requirements elicitation and early design**.

Key Notes

- UML use case diagrams are **trivial to learn** but require practice to write good use cases.
- Organizing use cases into relationships (*include*, *extend*) does not change system behavior, only the documentation.
- Use cases focus on **actor goals** and **observable results of value**.

Example UML Use Case Diagram (Conceptual)

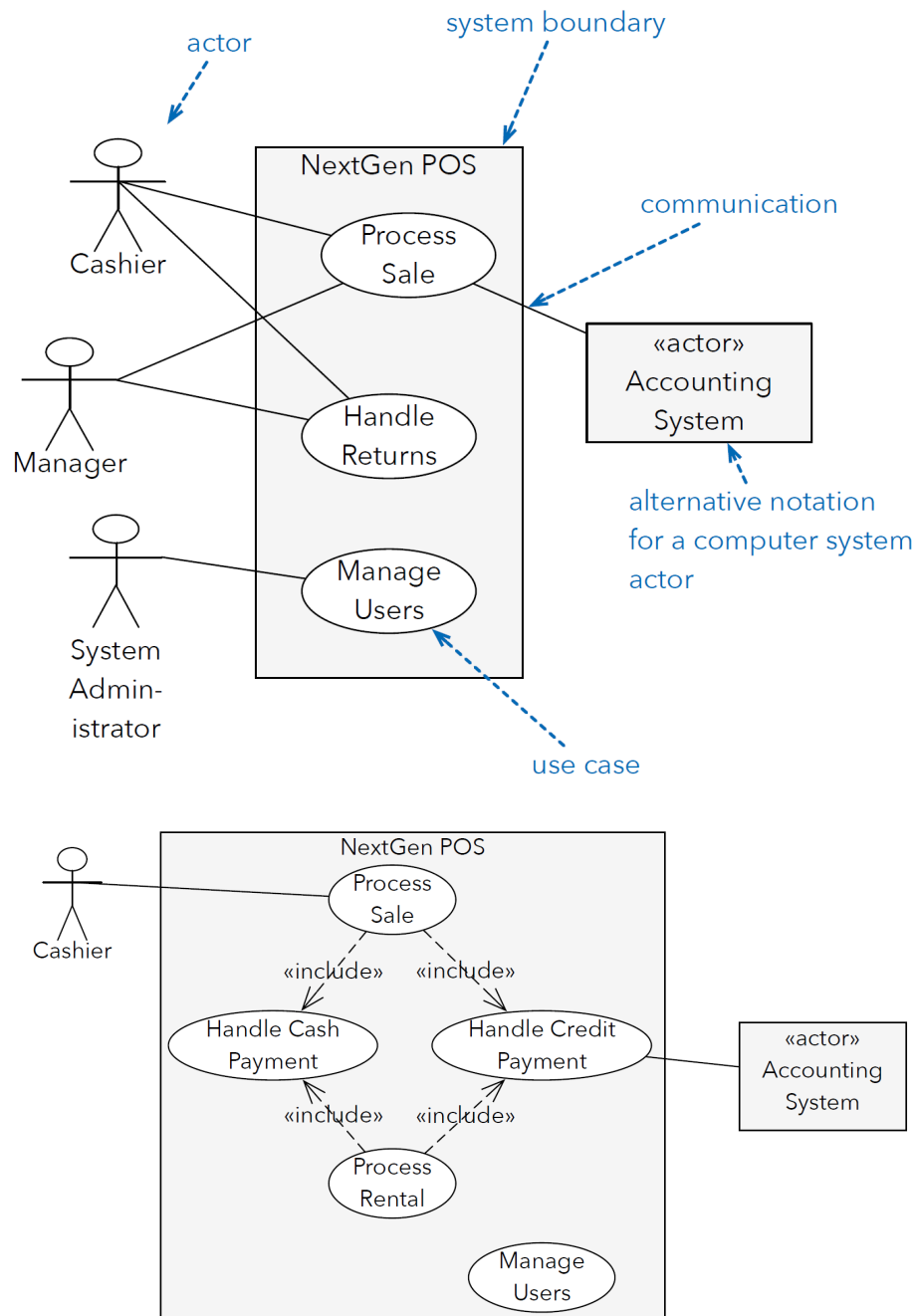


Figure 1: Conceptual UML Use Case Diagram for NextGen POS System

Symbols and Notations

- **Actor:** External entity interacting with the system (person, device, or other system).
- **Use Case:** Ellipse representing a goal the actor wants to achieve.
- **System Boundary:** Rectangle defining the scope of the system.

- **Association:** Solid line connecting actor to a use case.
- **Include / Extend:** Special relationships between use cases (optional in early analysis).